KHULNA UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science and Engineering

CSE 4122: Natural Language Processing Laboratory

Project Proposal

DeClickBait

Detecting and Correcting Clickbait Headlines Using Discriminative and Generative NLP

Submitted By

Jahid Hasan
Student ID: 2107064

Tasmir Hossain Zihad
Student ID: 2107071

Submitted To

Dr. K. M. Azharul Hasan
Professor

Md. Nazirul Hasan Shawon
Assistant Professor

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Abstract

Clickbait headlines exaggerate or misrepresent the content they point to, wasting reader attention and eroding trust in news and video platforms. **DeClickBait** is a natural language processing project that jointly (i) classifies whether a given title for a news article or YouTube video is clickbait, and (ii) generates a faithful, non-clickbait alternative title directly from the underlying content. Detection and correction are treated as complementary discriminative and generative tasks trained over the same content representation. A distinguishing aspect of this project is that the target corpus is naturally code-mixed: Bangladeshi digital media and YouTube content routinely blend Bangla, Banglish (Romanized Bangla), and English within the same title or article, and the system is designed around this reality rather than assuming a single clean language. The corpus combines established clickbait datasets with newly crawled and Gemini-assisted labeled data, and the project compares pretrained versus self-trained embeddings and a discriminative versus a generative modeling approach.

Introduction and Motivation

The rise of algorithmically ranked social feeds and video platforms has created strong commercial incentives for headline writers to prioritize engagement over accuracy. A clickbait title typically exploits a curiosity gap: it withholds key information or exaggerates the significance of the content to induce a click, often at the cost of misleading the reader or viewer. This is a well-studied problem for English news headlines [1] and social-media-shared article titles [2], and has more recently been studied for YouTube video titles [3] and for Bangla news headlines specifically [5]. Most existing work treats clickbait handling purely as a binary classification problem: given a title, predict whether it is clickbait. This flags content but does not help a reader who still does not know what the article or video is actually about. DeClickBait extends the problem: alongside classification, the system also generates a corrected, content-faithful title so the output is directly useful rather than just a flag.

A second motivation is linguistic. Bangladeshi online media, and especially YouTube titles and comments, are rarely written in a single, clean language. It is common to see Bangla script, Banglish (Bangla words typed in Roman script), and English mixed within one sentence or even one phrase. Any dataset built from real Bangladeshi sources therefore needs to be treated as inherently multilingual and code-mixed from the start, which shapes both the data collection procedure and the choice of embeddings described in later sections.

Objectives

- Build a system that takes a `(title, content)` pair, where content is either a news article body or a YouTube transcript, and predicts whether the title is clickbait.
- Generate a faithful, non-clickbait title directly from the content, independent of the classification decision.
- Handle a naturally code-mixed corpus containing Bangla, Banglish, and English text, rather than assuming a single language or script.
- Compare pretrained word embeddings against embeddings trained on the collected corpus, and report the effect on downstream performance.
- Implement and justify at least one discriminative classifier and one generative model, and compare them where feasible.
- Optionally explore unsupervised topic clustering as a secondary analysis.
- Document the full data collection and labeling methodology, since a meaningful portion of the corpus will be newly collected and LLM-assisted.

Related Work

Early clickbait detection relied on hand-engineered linguistic features combined with classical classifiers over headline-only datasets [1]. The Webis Clickbait Corpus 2017, built by crowdsourcing annotations for tweets from 27 major news publishers, pairs a shared post title with the linked article's original title and body text, and annotates clickbait strength on a continuous scale [2]; this structure closely matches the input and output format DeClickBait targets.

On the video side, Zannettou et al. combined manual annotation with channel-level heuristics to build a large labeled corpus of YouTube videos and associated metadata [3], while Lovejoy et al. released a large corpus of YouTube transcripts paired with titles for title generation and summarization, without clickbait labels [4].

For Bangla specifically, Mahtab et al. constructed the first Bangla clickbait dataset, consisting of over fifteen thousand expert-labeled news articles together with tens of thousands of unlabeled articles from clickbait-dense Bangla news sites [5]. This dataset is directly relevant to the Bangla portion of our corpus and to establishing what a "clickbait" pattern looks like in Bangla headlines specifically, rather than assuming English patterns transfer directly.

For embeddings, Word2Vec [7] and GloVe [8] remain standard pretrained baselines for English text. For Bangla and Bangla-English mixed text, BanglaBERT and its BanglishBERT variant provide transformer-based pretrained representations trained on large-scale Bangla and mixed Bangla-English corpora respectively [6], and are a natural pretrained option to compare against self-trained embeddings for the code-mixed portion of our corpus.

Proposed Methodology

System Overview

The system takes a `(title, content)` pair as input and produces two outputs: a clickbait label and a generated title. Both outputs are computed from a shared text representation, so the discriminative and generative components benefit from the same underlying understanding of the article or transcript, while remaining separately trainable and evaluable. Figure 1 shows the overall pipeline.

Dataset Construction

The corpus combines existing labeled datasets with newly crawled, Gemini-assisted labeled data, since no single existing dataset provides labeled clickbait status, full content, and a faithful reference title across the news, YouTube, and Bangla domains simultaneously.

Existing datasets. Table 1 summarizes the datasets identified for reuse, including a Bangla-specific corpus that directly informs the Bangla and Banglish portion of the project.

New data collection. News articles will be crawled from a mix of clickbait-leaning and neutral Bangladeshi and English-language outlets, recording title, body text, publication date, and source language mix. YouTube video metadata and transcripts will be collected via the YouTube Data API and a transcript-fetching tool, deliberately including Bangla-language and Banglish-titled channels alongside English channels, so the crawled data reflects the real code-mixed distribution rather than only clean English.

LLM-assisted labeling and field completion. Figure 2 shows the annotation pipeline. For newly crawled samples, and for existing dataset entries with missing fields, the Gemini API is used to generate a first-pass, silver-standard annotation: a preliminary clickbait label with a short justification, and a candidate faithful title generated from the content where no reference title exists. Gemini's multilingual capability makes it well suited to this step across Bangla, Banglish, and English text without needing separate pipelines per language. Both team members then manually review and edit a sample of this

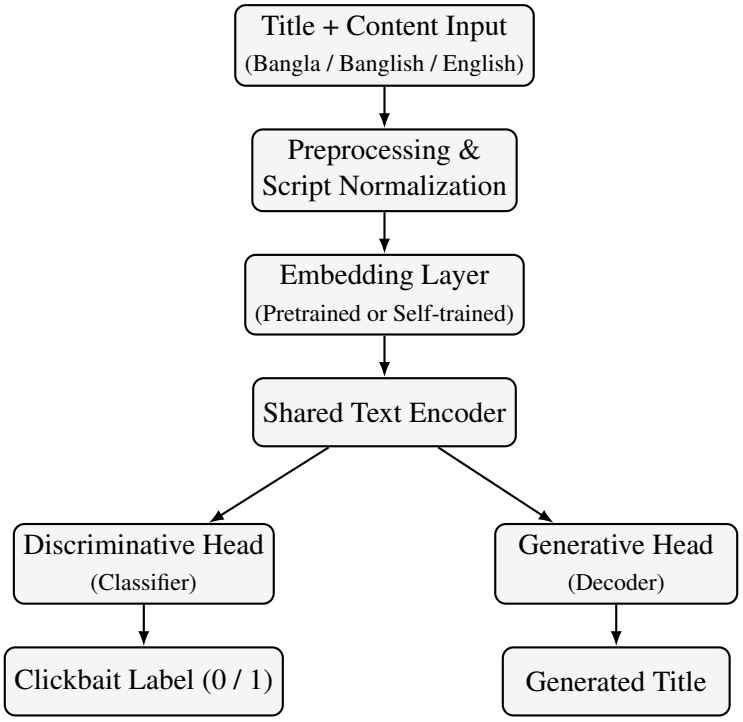


Figure 1: Overall system architecture. A shared encoder feeds a discriminative classification head and a generative title-decoding head.

output, correcting mislabeled instances, rewriting poor reference titles, and discarding unusable samples, before it is accepted into the final corpus.

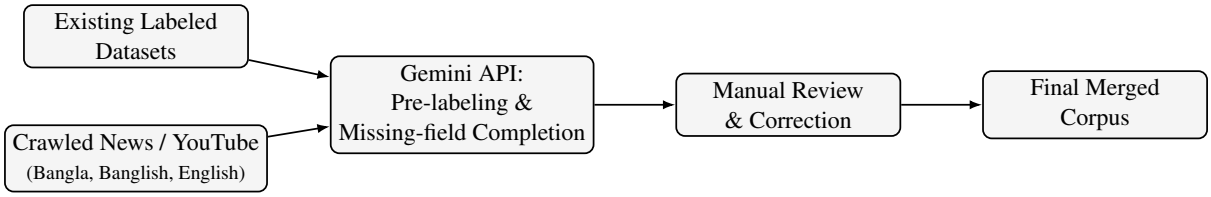


Figure 2: Dataset construction pipeline: existing and newly crawled data are pre-labeled by Gemini, then manually reviewed before entering the final corpus.

Cleaning and preprocessing. All samples are deduplicated, stripped of HTML and boilerplate, and checked for near-duplicate titles. Because the corpus is code-mixed, preprocessing also includes script identification (Bangla Unicode versus Roman script) and light normalization of Banglish spelling variants, rather than assuming a single tokenizer configuration is sufficient for the whole corpus.

Final data model. Each sample in the merged corpus follows the structure below.

```
Sample {
  id: string
  source_platform: enum { "news", "youtube" }
  language_mix: enum { "bangla", "banglish", "english", "mixed" }
  original_title: string        # the given, possibly clickbait title
  content_text: string          # article body or video transcript
  is_clickbait: int (0 | 1)
  actual_title: string          # faithful reference title
  label_source: enum { "existing_dataset", "llm_assisted_reviewed" }
  category: string (optional)   # topic tag, for clustering analysis
```

Dataset	Domain	Relevant fields
Webis Clickbait Corpus 2017 [2]	English news	Bait title, original article title, article body, clickbait strength score
Stop Clickbait Corpus [1]	English news headlines	Headline, binary clickbait label
BanglaBait Corpus [5]	Bangla news	Title, body, binary clickbait label, metadata
YouTube Titles and Transcripts [4]	YouTube (mostly English)	Title, full transcript, no clickbait label
YouTube Clickbait Classification (Kaggle)	YouTube	Title, engagement stats, binary clickbait label, no transcript

Table 1: Existing datasets to be reused, and the fields each one contributes.

```
  source_url: string
  collected_at: datetime
}
```

The `label_source` field records whether a sample's label and reference title came from an existing dataset or from the Gemini-assisted, human-reviewed pipeline, keeping every label traceable for the final report. The `language_mix` field lets later analysis check whether classification or generation quality differs across Bangla, Banglish, English, and mixed-language samples.

Embedding Strategy

Two embedding configurations are trained and compared under the same downstream architecture: pretrained embeddings, and embeddings trained from scratch on the merged corpus. Because the corpus spans Bangla, Banglish, and English, the pretrained option itself is not a single choice: Word2Vec or GloVe [7, 8] are considered for the English-only portion, while BanglaBERT and BanglishBERT [6] are considered for Bangla and Bangla-English mixed samples. As a lightweight intrinsic check of embedding quality before downstream training, cosine similarity between related and unrelated headline pairs is inspected:

$$\text{sim}(u, v) = \frac{u \cdot v}{\|u\| \, \|v\|} \tag{1}$$

where u and v are the embedding vectors of two title representations.

Model Architecture and Training Objectives

The discriminative component predicts the clickbait label from the shared content representation and is trained with the binary cross-entropy loss

$$\mathcal{L}_{cls} = -\frac{1}{N} \sum_{i=1}^{N} \Big[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \Big] \tag{2}$$

where $y_i \in \{0, 1\}$ is the ground-truth clickbait label for sample i and $\hat{y}_i$ is the predicted probability.

The generative component is a sequence-to-sequence model that produces a title token by token, conditioned on the content representation c, trained with the standard token-level negative log-likelihood

$$\mathcal{L}_{gen} = -\sum_{t=1}^{T} \log P(w_t \mid w_{<t}, c; \, \theta) \tag{3}$$

where w_t is the t-th token of the target title and θ denotes the model parameters. The two heads are trained and evaluated separately but share the embedding layer, allowing a controlled comparison of pretrained versus self-trained embeddings across both tasks.

Optional Unsupervised Exploration

As a secondary analysis, the corpus is clustered by topic (for example, k-means over document embeddings, or LDA) to examine whether clickbait tendency correlates with topic or with language mix. This is exploratory and does not gate the core supervised objectives.

Evaluation Plan

The classification component is evaluated with precision, recall, and F1-score,

$$P = \frac{TP}{TP + FP}, \qquad R = \frac{TP}{TP + FN}, \qquad F1 = \frac{2PR}{P + R} \tag{4}$$

with F1 prioritized given the likely class imbalance in real-world clickbait rates. The generation component is evaluated using ROUGE-L, based on the longest common subsequence (LCS) between a generated title X and a reference title Y of lengths n and m respectively,

$$R_{lcs} = \frac{LCS(X,Y)}{m}, \qquad P_{lcs} = \frac{LCS(X,Y)}{n}, \qquad F_{lcs} = \frac{(1+\beta^2)\, R_{lcs}\, P_{lcs}}{R_{lcs} + \beta^2 P_{lcs}} \tag{5}$$

supplemented by a small manual judgment pass to check whether generated titles are faithful to the content rather than merely fluent. Both metrics are reported separately for Bangla, Banglish, and English subsets, in addition to the pretrained versus self-trained embedding comparison.

Expected Outcomes and Deliverables

- A documented, merged, code-mixed dataset combining existing clickbait corpora with newly crawled, Gemini-assisted and manually verified Bangla, Banglish, and English samples.
- A trained clickbait classifier and a trained title-generation model, evaluated as described above.
- A comparative analysis of pretrained versus self-trained embeddings across languages.
- A simple input and output demonstration accepting a title and content in any of the three language forms and returning the predicted label and a suggested faithful title.
- A final report covering the data collection methodology, model architecture, results, and error analysis.

Conclusion

DeClickBait combines clickbait detection and title correction into a single project built around a shared content representation, giving concrete grounding for both a discriminative and a generative component as required by the course. The project also deliberately embraces the code-mixed reality of Bangladeshi digital media by building and documenting a Bangla, Banglish, and English corpus through a combination of existing datasets, targeted crawling, and Gemini-assisted labeling verified by manual review, rather than restricting the problem to clean, single-language English text.

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